Finite numerical-range certificates for polynomial matrix computations: a rational prototype with comparative controls

Robin Gounder*
Vaionex Corporation

Research draft, 7 October 2026

Abstract

A numerical-range inequality becomes a usable error certificate only after its continuous supremum and its numerical inputs have been bounded. We present a finite certificate for matrix-valued polynomials based on validated supporting half-planes, a boundary mesh, and coefficient norm bounds. The certificate is valid with the established complete spectral-set constant $1 + \sqrt{2}$; a separate hypothesis records the sharp constant 2 asserted in recent preprints and in the OpenAI mathematics release. We prove an explicit enclosure error estimate, distinguish eigenvalue validation from eigenpair residuals, and separate input uncertainty from evaluation roundoff. For Richardson residual powers, the classical numerical-radius power inequality already supplies an unconditional constant-two bound. The supremum over a polygon reduces exactly to its vertices, and a rational certificate can be checked without trigonometric functions or floating-point logarithms. Comparative controls show that enclosure geometry can lower a sufficient budget relative to a single-step induced-norm rate bound, while stronger exact-power certificates beat the prototype on its featured fixture. On the coupled 32×32 fixture at tolerance 10^{-8}, the numerical-range certificate gives 31 steps, compared with 11 for direct powers and 113 for the single-step norm-rate bound. The prototype supports real-rational scalar polynomials; the theorem permits matrix-valued polynomials. The experiments compare sufficient certificate budgets and construction costs. The contribution is a finite certificate specification, an explicit error decomposition, and reproducible diagnostics built from established spectral-set and numerical-range methods.

Keywords: numerical range; verified computation; matrix polynomial; spectral set; Richardson iteration; GMRES.
MSC 2020: 15A60, 65F10, 65G20, 47A25.

1 Introduction and scope

Polynomial computations occur in iterative linear solvers, matrix-function approximations, and stability analysis. For nonnormal matrices, evaluating a polynomial only at the eigenvalues can underestimate its operator norm. The numerical range retains directional information that can control this norm. A theorem of Crouzeix and Palencia gives, uniformly in the matrix and coefficient dimensions,

$$\|Q[A]\| \le (1 + \sqrt{2}) \max_{z \in W(A)} \|Q(z)\| \tag{1.1}$$

*Correspondence: robin@gounder.com.

for matrix-valued polynomials [1]. This complete bound permits coefficient matrices that do not commute. For an application, however, the right-hand side still contains a continuous set and a continuous maximization problem. An inaccurate outer enclosure or an uncertified eigenvalue can defeat the intended guarantee.

The numerical-range functional calculus builds on earlier operator integral representations [2], convex-domain spectral-set estimates [3], and uniform numerical-range bounds for polynomial functions [4]. Our finite arithmetic interface applies these established spectral-set estimates to finite, reproducible norm certificates.

The numerical determination of the field of values has a substantial history. Johnson developed its approximation using directional Hermitian eigenproblems [5]. Loisel and Maxwell developed a path-following algorithm, with error bounds for specified matrix classes and numerical cost–accuracy comparisons [6]. Field-of-values computation is also available in existing scientific software, for example the Chebfun exposition and implementation [7]. These methods supply the supporting-line geometry used in the present certificate.

Numerical-range estimates already support matrix-function algorithms. Beckermann and Reichel connect Faber approximation and matrix-function error estimates [8]. Crouzeix and Greenbaum give spectral-set estimates with applications to GMRES and rational Krylov methods [9]. Our exact-arithmetic GMRES corollary connects the finite certificate with the standard residual-minimization property of the method [10].

The polynomial-approximation viewpoint also underlies geometric analyses of Krylov methods [11]. Beckermann obtained numerical-range GMRES estimates through Faber polynomials [12]. Eigenvalues alone cannot prescribe a nonnormal GMRES convergence history: Greenbaum, Pták and Strakoš constructed any prescribed nonincreasing positive residual curve with any chosen nonzero spectrum [13].

The sharp constant is also the subject of literature preceding the repository studied here. Lorist and Schwenninger assert a general scalar constant-two proof in a 2026 preprint [14]. Crouzeix and Greenbaum subsequently use that result to update spectral-set estimates [15]. Complete constant-two results have been reported for weighted shifts [16] and for matrices of order at most three [17]. The OpenAI release asserts the complete inequality in arbitrary finite dimensions [18]. Scalar constant-two applications therefore have a literature basis preceding the repository release.

This paper addresses a narrower operational question: what finite evidence suffices to turn a numerical-range bound into a reproducible norm certificate? Its contributions are an explicit finite-input theorem, an error decomposition that can be audited separately, and an exact rational realization for scalar residual polynomials. The geometric ingredients and the spectral-set applications are established methods. The controls below test recomputability and quantify conservatism through comparisons with competing certificates, including norms of matrix powers.

Matrix-function theory and conditioning are treated systematically by Higham [19]. Exponential integrators provide a concrete setting in which matrix functions applied to vectors are required [20]; truncated Taylor matrix polynomials are used in the exponential-action algorithm of Al-Mohy and Higham [21]. These existing applications motivate future certificate work. The present prototype implements scalar polynomial and Richardson certificates; exponential-action algorithms provide a further application setting.

Evidence status. We fix the repository snapshot at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, dated 6 October 2026, and record the source audit as of 7 October 2026 [22]. Its scope document describes a sharp complete constant-two result and links several Lean comparator statements [23]. The source assessment uses the supplied scope document, comparator configuration,

and manuscript at this snapshot. This evidence describes the reported theorem scope and its comparison statements. For arbitrary polynomials, unconditional conclusions below use $1+\sqrt{2}$, and the general constant-two version is stated under an explicit hypothesis. Fixed Richardson powers instead use the classical unconditional numerical-radius power inequality, as explained in Section 6. This distinction separates the theorem assumptions from the finite certificate arithmetic.

2 Notation and the theorem interface

Let $A \in \mathbb{C}^{n\times n}$, where $n \geq 1$. All unsubscripted vector and operator norms are Euclidean norms; the subscripts 1 and ∞ denote the usual induced norms. Write

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n,\ \|x\| = 1\}.$$

In finite dimensions the numerical range is nonempty, compact, and convex. For $c \in \mathbb{C}$, let $T = A - cI_n$, and write

$$Q(z) = \sum_{k=0}^{d} B_k(z-c)^k, \qquad B_k \in \mathbb{C}^{s\times s}, \quad s \geq 1.$$

Its operator evaluation is

$$Q[A] = \sum_{k=0}^{d} T^k \otimes B_k. \tag{2.1}$$

The matrix space is the first tensor factor; interchanging the factors is unitarily equivalent but should not be done silently in software. Coefficient products are not assumed to commute.

Definition 2.1 (Admissible constant)**.** A constant C is admissible for the computations considered here if

$$\|Q[A]\| \leq C \max_{z\in W(A)} \|Q(z)\|$$

holds for every stated matrix and coefficient size and polynomial. The established universal complete choice is $C = 1+\sqrt{2}$ [1]. The hypothesis $\mathcal{H}_2$ is that the same complete statement holds universally with $C = 2$.

We use $\mathcal{H}_2$ only to label conclusions that require it. For scalar coefficients, the scalar constant-two statement suffices; the complete hypothesis is stronger than necessary. Conversely, a scalar theorem alone does not justify an arbitrary matrix-coefficient certificate. For a normal A, the spectral theorem supplies $C = 1$ directly:

$$\|Q[A]\| = \max_{\lambda\in\operatorname{spec}(A)} \|Q(\lambda)\| \leq \max_{z\in W(A)} \|Q(z)\|.$$

A producer should use such a justified stronger bound when it recognizes the required structure.

A *validated* quantity in this paper is a proved upper bound for the specified exact mathematical input. It may be obtained by rational arithmetic, outward interval arithmetic, or another justified method. A decimal approximation, an unrecorded numerical tolerance, and a plot are not certificates. The distinction follows the general philosophy of verification methods surveyed by Rump [24].

3 Finite outer enclosures

3.1 Directional support and validated producers

For $\theta \in \mathbb{R}$, set

$$H_\theta(T) = \frac{e^{-i\theta}T + e^{i\theta}T^*}{2}, \qquad h(\theta) = \lambda_{\max}(H_\theta(T)).$$

For every unit vector x,

$$\operatorname{Re}(e^{-i\theta}x^*Tx) = x^*H_\theta(T)x.$$

The Rayleigh–Ritz principle identifies $h(\theta)$ with the support function of $W(T)$.

Choose directions $\theta_1, \dots, \theta_J$ in cyclic order modulo 2π, and validated numbers $\bar{h}_j \geq h(\theta_j)$. Define

$$P = c + \bigcap_{j=1}^{J} \{w \in \mathbb{C} : \operatorname{Re}(e^{-i\theta_j}w) \leq \bar{h}_j\}. \tag{3.1}$$

This is an outer enclosure. To use its vertices or boundary, its nonemptiness and boundedness must also be established.

For exact rational geometry, unit normals are unnecessary. Put

$$S = \frac{T + T^*}{2}, \qquad J_T = \frac{T - T^*}{2i}.$$

Given a nonzero real normal $q = (a, b)$, the relevant Hermitian matrix is $H_q = aS + bJ_T$, and

$$\sup_{z \in W(T)} (a \operatorname{Re} z + b \operatorname{Im} z) = \lambda_{\max}(H_q).$$

A validated upper bound g_q gives the half-plane $a \operatorname{Re}(z - c) + b \operatorname{Im}(z - c) \leq g_q$. With rational a, b, g_q, c, intersections and feasibility checks can be performed exactly. There is no need to treat rounded sine and cosine values as exact unit normals.

An elementary valid producer uses the Gershgorin row bound

$$g_q = \max_i \left((H_q)_{ii} + \sum_{\ell \neq i} u_{i\ell} \right), \qquad u_{i\ell} \geq |(H_q)_{i\ell}|. \tag{3.2}$$

Indeed, for an eigenpair choose a component of maximal modulus. Its eigenvalue equation gives $\lambda \leq (H_q)_{ii} + \sum_{\ell \neq i} |(H_q)_{i\ell}|$, because the eigenvalue is real. This proves the required upper bound. The producer may be conservative; validity does not imply accuracy.

Remark 3.1 (Eigenpair residuals do not certify the upper edge). For $H = \operatorname{diag}(0, 100)$, the unit vector $v = (1, 0)^T$ has Rayleigh quotient 0 and eigenpair residual 0, while $\lambda_{\max}(H) = 100$. A single small Ritz residual cannot justify an upper support line. It can assist in a lower bound, or be combined with additional certified spectral information. Other valid upper producers include a verified positive-semidefinite test for $gI - H$ and a validated full decomposition with controlled orthogonality and remainder. For a strict upper support, one can instead verify positive definiteness of $gI - H_q$. Rump develops rigorous Cholesky-based sufficient tests for Hermitian positive definiteness with rounding errors included [25]. Such a validator is a possible alternative support producer; it is not supplied by the present implementation.

3.2 An explicit enclosure error

A convergence statement requires more than upper supports. In particular, refining the directions of a Gershgorin construction need not remove its spectral slack.

Proposition 3.2 (Radius and enclosure excess)**.** *Assume $W(T) \subseteq \{z : |z| \le R\}$, where $R \ge 0$, and let the maximum cyclic gap between the unit-normal directions be $\Delta < \pi$. Suppose, for some $\eta \ge 0$,*

$$h(\theta_j) \le \bar{h}_j \le h(\theta_j) + \eta \quad \text{for every } j.$$

Then P in (3.1) is a nonempty compact convex outer enclosure, every $z \in P$ satisfies

$$|z - c| \le \rho, \qquad \rho = (R + \eta)\sec(\Delta/2),$$

and its one-sided Hausdorff excess satisfies

$$\sup_{z \in P} \operatorname{dist}(z, W(A)) \le d_P := \eta + 2(R + \rho)\sin(\Delta/4). \tag{3.3}$$

Proof. The support identity implies $W(A) \subseteq P$, proving nonemptiness. For any nonzero $w = z - c$ in the intersection, select a sampled normal within angular distance $\Delta/2$ of $w/|w|$. Then

$$|w|\cos(\Delta/2) \le \operatorname{Re}(e^{-i\theta_j} w) \le \bar{h}_j \le R + \eta.$$

This establishes boundedness and the radius estimate. The intersection is closed and convex, hence compact.

Work in shifted coordinates and write $K = W(T)$. For $w \in P - c$ outside K, let y be its Euclidean projection onto K. The unit vector $u = (w - y)/|w - y|$, identified with a complex unit normal, supports K at y. Hence $\operatorname{dist}(w, K) = u \cdot w - h_K(u)$. Choose a sampled unit normal u_j within angle $\Delta/2$ of u. The support function is R-Lipschitz as a function of the unit normal:

$$|h_K(u) - h_K(u_j)| \le R|u - u_j|.$$

The sampled inequality and $|w| \le \rho$ give

$$u \cdot w - h_K(u) \le \eta + (R + \rho)|u - u_j| \le \eta + 2(R + \rho)\sin(\Delta/4).$$

Points in K have distance zero, so the same bound holds throughout. □

The estimate gives an explicit upper budget in terms of support slack and angular resolution. Any validated disk bound may serve as R; for instance, $R \ge \|T\|$ is sufficient. A useful bound on η can be obtained from a certified lower support value $\underline{h}_j$: take $\eta = \max_j(\bar{h}_j - \underline{h}_j)$. An exact Rayleigh quotient supplies such a lower value. The difference must include Gershgorin overestimation as well as arithmetic rounding. Merely setting η equal to an eigensolver tolerance is unjustified.

If input uncertainty satisfies $\|T - \widehat{T}\| \le \varepsilon_A$, a support bound for $\widehat{T}$ can be safely enlarged by $|q|\varepsilon_A$, since $\|H_q(T) - H_q(\widehat{T})\| \le |q|\varepsilon_A$. This certifies an enclosure of the uncertain input itself. The alternative nominal-input transfer in Section 5 should not be combined with this enlargement in a way that double-counts the same error.

4 The finite polynomial certificate

Let P be a nonempty compact convex polygon containing $W(A)$. For a polygon with nonempty interior, let $Z = \{z_1, \dots, z_L\}$ be a finite subset of its boundary such that every boundary point is within distance δ of Z. If P is a segment or point, require instead that every point of P be within distance δ of Z. This convention avoids invoking a two-dimensional maximum principle on a degenerate domain.

Choose validated quantities

$$\rho \geq \max_{z \in P} |z - c|, \quad \beta_k \geq \|B_k\|, \quad u_\ell \geq \|Q(z_\ell)\|,$$

and put

$$D_Q = \sum_{k=1}^{d} k \beta_k \rho^{k-1}, \qquad U_Q = \max_{\ell} u_\ell + \delta D_Q. \tag{4.1}$$

The empty derivative sum is zero. Zeroth powers are interpreted as one, including when $\rho = 0$.

Theorem 4.1 (Finite-input certificate)**.** *Under the stated validated-input conditions,*

$$\max_{z \in P} \|Q(z)\| \leq U_Q.$$

Consequently,

$$\|Q[A]\| \leq C U_Q \tag{4.2}$$

for every admissible constant C. The certificate is unconditional for $C = 1 + \sqrt{2}$, and its $C = 2$ version follows under $\mathcal{H}_2$. A rational upper bound $C_{\mathrm{up}} \geq C$ may replace C in a numerical certificate.

Proof. On the disk $|z - c| \leq \rho$, $\|Q'(z)\| \leq D_Q$. Integrating along a straight line segment in that convex disk shows that $\|Q(z) - Q(w)\| \leq D_Q |z - w|$. The mesh therefore bounds each covered point by U_Q.

When P has interior, for any fixed unit vectors $a, b \in \mathbb{C}^s$, the scalar polynomial $a^* Q(z) b$ satisfies the maximum-modulus principle on P. For any $z \in P$,

$$|a^* Q(z) b| \leq \max_{w \in \partial P} \|Q(w)\|.$$

Taking the supremum over a, b bounds $\|Q(z)\|$ by the boundary supremum, which the mesh estimate bounds by U_Q. For a degenerate P, the mesh already covers the entire domain. Finally, $W(A) \subseteq P$ and the admissible spectral-set inequality give (4.2). □

Proposition 4.2 (Auditable overestimation budget)**.** *Suppose Proposition 3.2 applies and the sample bounds also satisfy $u_\ell \leq \|Q(z_\ell)\| + \eta_{\mathrm{sample}}$, with a known $\eta_{\mathrm{sample}} \geq 0$. If $M_Q = \max_{z \in W(A)} \|Q(z)\|$, then*

$$U_Q \leq M_Q + D_Q(d_P + \delta) + \eta_{\mathrm{sample}}. \tag{4.3}$$

Proof. For each sample point choose a closest point in the compact set $W(A)$. Both points lie in the radius-ρ disk. The Lipschitz estimate gives $\|Q(z_\ell)\| \leq M_Q + D_Q d_P$. Adding the sample slack and mesh term proves the claim. □

Theorem 4.1 needs only the upper bounds. The two-sided accuracy assumptions in Proposition 4.2 quantify conservatism; they are not needed to establish validity. Coefficient norm bounds may use a validated Frobenius norm or $\sqrt{\|B_k\|_1 \|B_k\|_\infty}$. These inexpensive choices can introduce substantial slack for matrix-valued coefficients.

Finite construction. A checker can recompute the supporting inequalities, enumerate the polygon vertices, bound the enclosing radius, and subdivide each edge. If an edge of length ℓ is split into m equal subsegments and both endpoints are sampled, every point of that edge is within $\ell/(2m)$ of a sample. A validated upper bound on ℓ gives a valid mesh radius. The point evaluations, coefficient bounds, and all final arithmetic must be validated. A producer that checks only the sampled function values has not bounded the values between samples.

5 Input uncertainty and evaluation roundoff

A geometric certificate bounds an exact operator. If the input is uncertain or an implementation evaluates a nearby polynomial, those errors require their own evidence.

Proposition 5.1 (Nominal-input transfer)**.** *Let* $Q(z) = \sum_{k=0}^{d} B_k(z-c)^k$ *and* $\widehat{Q}(z) = \sum_{k=0}^{d} \widehat{B}_k(z-c)^k$*. Suppose*

$$\|A - \widehat{A}\| \le \varepsilon_A, \qquad \|B_k - \widehat{B}_k\| \le \varepsilon_k, \qquad \|B_k\| \le \beta_k,$$

and $\max\{\|A - cI\|, \|\widehat{A} - cI\|\} \le M$*. Then*

$$\|Q[A] - \widehat{Q}[\widehat{A}]\| \le E_{\text{input}} := \varepsilon_A \sum_{k=1}^{d} k\beta_k M^{k-1} + \sum_{k=0}^{d} \varepsilon_k M^k. \tag{5.1}$$

If a finite certificate bounds $\|\widehat{Q}[\widehat{A}]\| \le C\widehat{U}$*, it follows that* $\|Q[A]\| \le C\widehat{U} + E_{\text{input}}$*.*

Proof. Write $T = A - cI$ and $\widehat{T} = \widehat{A} - cI$. The telescoping identity

$$T^k - \widehat{T}^k = \sum_{j=0}^{k-1} T^{k-1-j}(T - \widehat{T})\widehat{T}^j$$

holds without commutativity and gives $\|T^k - \widehat{T}^k\| \le kM^{k-1}\varepsilon_A$. Split each term of the difference as

$$T^k \otimes B_k - \widehat{T}^k \otimes \widehat{B}_k = (T^k - \widehat{T}^k) \otimes B_k + \widehat{T}^k \otimes (B_k - \widehat{B}_k).$$

Use the product rule for the spectral norm of a Kronecker product and the triangle inequality. □

If an implementation produces Y with a validated bound $\|Y - \widehat{Q}[\widehat{A}]\| \le E_{\text{alg}}$, then

$$\|Y\| \le C\widehat{U} + E_{\text{alg}}, \qquad \|Y - Q[A]\| \le E_{\text{alg}} + E_{\text{input}}.$$

These statements distinguish the computed output norm from its distance to the intended operator. The value E_{alg} requires validation for the evaluation algorithm. The implementation below supplies exact-rational geometry; floating-point Horner and Krylov evaluation require their corresponding roundoff analyses. This separation is consistent with the forward/backward error and conditioning distinctions in Higham [26, Chapters 1–3 and 5]. An exact polynomial norm certificate therefore does not by itself provide a floating-point evaluation-error bound.

A similar separation applies to matrix-function approximation. If f is holomorphic on a neighborhood of P and $\max_{z\in P} |f(z) - p(z)| \le \varepsilon_{\text{approx}}$, the established functional-calculus bound gives

$$\|f(A) - p(A)\| \le (1 + \sqrt{2})\varepsilon_{\text{approx}}.$$

For a computed approximation Y, a validated evaluation error $\|Y - p(A)\| \le E_{\text{alg}}$ gives total error $(1 + \sqrt{2})\varepsilon_{\text{approx}} + E_{\text{alg}}$. This extends the theorem interface to matrix-function approximation. The scalar reference implementation supplies polynomial mesh bounds; approximation and evaluation-error bounds remain separate validated inputs.

6 Residual polynomials and practical decisions

6.1 A vertex certificate for Richardson iteration

For $Ax = b$, fixed-step Richardson iteration with parameter $\alpha \in \mathbb{C}$ has exact residual

$$r_m = (I - \alpha A)^m r_0.$$

Let the enclosing polygon P have vertex set V, including endpoints when P is a segment and its single point when it is a point.

Lemma 6.1 (Vertex reduction)**.** *Let $q_P = \max_{v \in V} |1 - \alpha v|$. For every integer $m \geq 1$,*

$$\max_{z \in P} |(1 - \alpha z)^m| = q_P^m.$$

Proof. Every point of $P = \operatorname{conv}(V)$ is a convex combination $z = \sum_v t_v v$, with $t_v \geq 0$ and $\sum_v t_v = 1$. The triangle inequality gives

$$|1 - \alpha z| = \left| \sum_v t_v (1 - \alpha v) \right| \leq \sum_v t_v |1 - \alpha v| \leq q_P.$$

The bound is attained at a maximizing vertex. Raising to the positive integer power m proves the result. □

Define the numerical radius by $w(B) = \max_{z \in W(B)} |z|$, using complex unit vectors even when B is real. The classical numerical-radius power inequality gives $w(B^m) \leq w(B)^m$; together with $\|B\| \leq 2w(B)$, this yields $\|B^m\| \leq 2w(B)^m$ [27, Section 3.2]; see also Lewis and Overton [28, Eq. (1.1)]. These established facts give a sharper unconditional bound for this particular polynomial family than the general Crouzeix–Palencia constant.

Proposition 6.2 (Classical power certificate)**.** *For $E = I - \alpha A$ and every integer $m \geq 1$,*

$$\|E^m\| \leq 2w(E)^m \leq 2q_P^m.$$

No constant-two Crouzeix hypothesis is needed.

Proof. The affine numerical-range identity gives $W(E) = 1 - \alpha W(A)$, so $w(E) \leq q_P$. Apply the power inequality and $\|E^m\| \leq 2w(E^m)$. □

Corollary 6.3 (Residual budget)**.** *For every $m \geq 1$, the unconditional choice $C = 2$ gives the following bound. The general spectral-set choice $C = 1 + \sqrt{2}$ remains a valid, weaker comparator:*

$$\|r_m\| \leq C q_P^m \|r_0\|.$$

If $0 < q_P < 1$ and $0 < \tau < 1$, a sufficient exact-arithmetic iteration budget for $\|r_m\|/\|r_0\| \leq \tau$, with $r_0 \neq 0$, is

$$m_C = \left\lceil \frac{\log(\tau/C)}{\log q_P} \right\rceil. \tag{6.1}$$

If $q_P = 0$, one step suffices. If $q_P \geq 1$, this certificate does not guarantee eventual contraction.

The constant two here follows from Proposition 6.2, even if the general hypothesis $\mathcal{H}_2$ is unresolved. This distinction is essential: the Richardson certificate follows from the classical power inequality. Geometric contractivity is a sufficient certificate condition; the iteration can converge even when that condition fails.

For real rational α and rational polygon vertices,

$$q_P^2 = \max_{v=x+iy\in V}((1-\alpha x)^2 + \alpha^2 y^2)$$

is rational. A checker can avoid logarithms and square roots in the stopping decision by testing

$$C_{\mathrm{up}}^2 (q_P^2)^m \le \tau^2. \tag{6.2}$$

Testing successive positive integers until this inequality holds gives the smallest budget certified by the chosen rational constant upper bound. A configured search cap is a software resource limit, not evidence that no certificate exists.

6.2 GMRES and inexact recurrence

In exact arithmetic, the m-step GMRES residual minimizes $\|p(A)r_0\|$ over polynomials of degree at most m with $p(0) = 1$ [10]. The Richardson polynomial is an admissible comparator. With $C = 2$ unconditionally by Proposition 6.2,

$$\|r_m^{\mathrm{GMRES}}\| \le C q_P^m \|r_0\|.$$

This bounds the mathematical minimal residual in exact arithmetic. Polynomial selection and finite-precision execution require the corresponding GMRES algorithm analysis.

For a perturbed residual recurrence,

$$\widetilde{r}_{j+1} = (I - \alpha A)\widetilde{r}_j + \xi_j, \qquad \|\xi_j\| \le \varepsilon_j,$$

take $C = 2$ unconditionally, define $\kappa_0 = 1$ and $\kappa_k = C q_P^k$ for $k \ge 1$. Unrolling the recurrence proves

$$\|\widetilde{r}_m\| \le C q_P^m \|\widetilde{r}_0\| + \sum_{j=0}^{m-1} \kappa_{m-1-j}\varepsilon_j, \qquad m \ge 1. \tag{6.3}$$

No independence or cancellation of the perturbations is assumed. If $q_P < 1$ and every $\varepsilon_j \le \varepsilon$, this gives

$$\|\widetilde{r}_m\| \le C q_P^m \|\widetilde{r}_0\| + \varepsilon \left(1 + C q_P \frac{1 - q_P^{m-1}}{1 - q_P}\right).$$

The quantity $\varepsilon(1 + Cq_P/(1-q_P))$ is a conservative asymptotic upper bound for the perturbation contribution. It is not a lower bound on attainable residuals: the permitted perturbations may all be zero. Deriving the actual ε_j, and relating recursive residuals to explicitly recomputed residuals, remain necessary tasks for a floating-point solver. Inexact Krylov methods have their own analysis of controlled matrix-vector inaccuracies [29]. That theory provides algorithm-specific perturbation budgets that can be used with the present recurrence estimate.

6.3 When the constant can matter

For the same validated U_Q, replacing $1+\sqrt{2}$ by 2 reduces the multiplicative bound by

$$1-\frac{2}{1+\sqrt{2}}=3-2\sqrt{2}\approx 0.1716.$$

For Richardson powers this replacement is already justified unconditionally by Proposition 6.2; for a general polynomial, a constant-two spectral-set conclusion still has its separate theorem obligation. It does not change the exponential rate q_P. For $0<q_P<1$, the budgets in (6.1) obey

$$0\le m_{1+\sqrt{2}}-m_2\le\left\lceil\frac{\log((1+\sqrt{2})/2)}{-\log q_P}\right\rceil.$$

The difference may be zero. These budgets refer to exact constants and exact q_P; outward numerical bounds can add slack.

The enclosure budget is often more significant. If $M_Q>0$ and $U_Q=M_Q(1+e)$, then the constant-two bound beats the ideal established bound $(1+\sqrt{2})M_Q$ exactly when

$$e<\frac{\sqrt{2}-1}{2}\approx 0.2071.$$

This comparison is a diagnostic against an ideal reference, not an observable acceptance test unless M_Q is also known. It explains why a large enclosure excess can erase the advantage of a sharper constant. An additive algorithm error can further dominate both bounds.

The alternative induced-norm certificate should always be reported. For $E=I-\alpha A$,

$$\|E^m\|\le\min\left\{\|E\|^m,\left(\|E\|_1\|E\|_\infty\right)^{m/2}\right\}.$$

A validated upper bound on either rate gives a competing budget. Whenever both it and Cq_P^m are available, their minimum is a valid and at least as informative certificate. The single-step rate is only a basic baseline. It can lose substantial information that remains in E^m, so beating this baseline is insufficient evidence of superiority. Scalable norm estimation also requires care in a validated comparison: the block one-norm estimator of Higham and Tisseur returns a lower bound, not an outward upper certificate [30, Section 5]. A diagnostic estimate must acquire separate upper-bound validation before it can replace c_j in the certificate below.

Proposition 6.4 (Standard matrix-power comparison)**.** *Let $k\ge 1$, let $c_0=1$, and suppose that $c_j\ge\|E^j\|^2$ for $1\le j\le k$. For integers $m=qk+r$, with $q\ge 0$ and $0\le r<k$,*

$$\|E^m\|^2\le c_k^q c_r.$$

In particular, an exactly computed real rational matrix power gives the validated choice $c_j=\|E^j\|_1\|E^j\|_\infty$.

Proof. The identity $E^{qk+r}=(E^k)^qE^r$ and submultiplicativity give $\|E^{qk+r}\|\le\|E^k\|^q\|E^r\|$. Square this inequality and substitute the stated upper bounds. The induced-norm product bounds the squared Euclidean operator norm. □

This standard comparison supplies the matrix-power baseline. For an exact rational power, the target test $c_m\le\tau^2$ requires no spectral-set constant or square root. Scanning successive powers can yield a stronger sufficient budget than both a single-step rate and a numerical-range enclosure. Direct powers and block-power reuse have their own arithmetic costs, including fraction growth. A performance comparison must account for those costs rather than rank methods by iteration budgets alone. A practical comparison therefore reports bound strength, construction costs, and reuse of computed powers.

7 Reference realization and analytic controls

7.1 A rational producer and a recomputation contract

For a real rational matrix A, write $S = (A + A^T)/2$ and $K = (A - A^T)/2$. The support matrix for $q = (a, b)$ is

$$H_q = aS - ibK.$$

Its diagonal entries are aA_{ii}, and an off-diagonal modulus is $\sqrt{a^2 S_{i\ell}^2 + b^2 K_{i\ell}^2}$. A rational upper square root gives a fully rational Gershgorin row witness. At precision b_{sq}, an integer square-root routine can find the smallest integer $t \geq 0$ for which

$$t^2 q_{\mathrm{den}} \geq p\, 2^{2b_{\mathrm{sq}}},$$

where $p/q_{\mathrm{den}} \geq 0$ is the radicand. Then $t/2^{b_{\mathrm{sq}}}$ is an outward upper bound, with excess less than $2^{-b_{\mathrm{sq}}}$; exact dyadic roots have zero excess. This arithmetic statement does not bound the separate Gershgorin slack.

One direction set consists of the integer points on the boundary of $[-k, k]^2$, $k \geq 1$, with duplicate corners removed. It contains $8k$ directions and the four axes. Axis constraints prove boundedness. All pairwise nonparallel line intersections are formed rationally and tested against every inequality. Their feasible convex hull produces the bounded polygon, including segment and point cases. This elementary enumeration makes each feasible intersection auditable and suits the dimensions and direction counts of the prototype.

A certificate should record the exact input entries and α, the normal set, all row witnesses, all support bounds, feasible vertices, q_P^2, the theorem mode, the rational upper constant, and the stopping inequality. A checker must recompute these quantities from the input. It should reject inconsistencies between the declared input and its numerical evidence, including an inward support, a missing extremal vertex, an understated squared radius, or a false stopping inequality. The caller must compare the declared matrix with the intended input before relying on the result. Rational data should be serialized as exact numerator–denominator pairs or unambiguous rational strings, not rounded JSON numbers.

The checker recomputes the arithmetic evidence for the stated exact input and theorem mode. The implementation supports scalar polynomials with real rational coefficients and Richardson polynomials with positive real rational step parameters, for real rational input matrices. Application use requires validation of the model that supplied A and of the solver evaluation. The general matrix-valued mesh theorem specifies an extension to complex inputs and matrix coefficients, with coefficient and point-norm validators as the required additional components.

7.2 Normal and sharp controls

For $A = \mathrm{diag}(1, 2)$ and $\alpha = 2/3$, the numerical range is $[1, 2]$ and $q_P = 1/3$ for the exact segment enclosure. The true residual norm is 3^{-m}. The spectral theorem gives the stronger constant 1. A universal-C result provides a consistency check; the spectral bound is stronger for this matrix.

For

$$J = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix},$$

the numerical range is the closed unit disk and $\|J\| = 2$. The scalar polynomial $p(z) = z$ has ratio 2. This is the standard sharpness control; the low-dimensional and weighted-shift setting was studied before this release [16]. On $I + J$, Richardson with $\alpha = 1$ has $q = 1$ on the true numerical range

and no strict contraction certificate of the stated form, even though the residual is zero after two exact steps. Failure to certify contraction must not be reported as failure of the iteration.

Let N_n be the nilpotent upper shift, with ones on the first superdiagonal. For $A = I_n + tN_n$, $t > 0$, and $\alpha = 1$,

$$(I - A)^m = (-t)^m N_n^m, \qquad \|(I - A)^m\| = \begin{cases} t^m, & 0 \le m < n, \\ 0, & m \ge n. \end{cases}$$

For $n \ge 2$, the induced 1- and infinity-norm rate is t; for $n = 1$ it is zero. On the displayed $n = 32, t = 1/5$ control, the Gershgorin certificate needs 13 steps versus 12 for the one-step norm bound. This is not a uniform ordering over the family: for $n = 2, t = 99/100$, eight directions with 40-bit roots certify 31 steps versus 1,833 from the one-step norm rate, while direct powers certify two steps. Certificate validity and a selected-baseline improvement are assessed together with the competing exact-power budget.

7.3 A coupled nonnormal family

For even $n \ge 4$, define the real upper-bidiagonal matrix by

$$A_{ii} = \begin{cases} 1, & i \text{ odd}, \\ 11/10, & i \text{ even}, \end{cases} \qquad A_{i,i+1} = \begin{cases} -4/5, & i \text{ odd}, \\ -1/100, & i \text{ even}, \end{cases}$$

and take $\alpha = 1$. The nonzero weak links join the two-dimensional blocks into a coupled nonnormal matrix. The displayed rational entries define the analytic fixture.

For $E = I - A$, direct row and column sums give

$$\|E\|_\infty = 4/5, \qquad \|E\|_1 = 9/10, \qquad q^2_{\text{induced}} = 18/25.$$

For the real-axis supports of A, the largest off-diagonal Gershgorin row radius is $r = (4/5{+}1/100)/2 = 81/200$. The four axis supports alone give the valid rectangle

$$P_{\text{box}} = [\,1 - r,\ 11/10 + r\,] + i[-r, r].$$

Its Richardson vertex value is exactly

$$q^2_{\text{box}} = (1/10 + r)^2 + r^2 = \frac{8381}{20000} < \frac{18}{25} < 1. \tag{7.1}$$

These statements use rational arithmetic and the proved row bound. They hold independently of $\mathcal{H}_2$. For a relative target $\tau = 10^{-8}$, (6.2) with $C_{\text{up}} = 17/7$ gives a sufficient box budget of 45 iterations, while the single-step induced-norm rate budget is 113. Here $17/7 \ge 1 + \sqrt{2}$, since $(10/7)^2 > 2$. The inequalities were checked exactly; the corresponding 44- and 112-step tests fail. These are conservative sufficient budgets for the exact residual bound.

For $n = 32$, $k = 1$, and $b_{\text{sq}} = 40$, the eight-direction prototype gives

$$q_P^2 = \frac{1069775589843180464514493}{377789318629571617095680 0} \approx 0.283167240864243. \tag{7.2}$$

Thus $q_P \approx 0.532134607843018$. The polygon has eight vertices; one maximizing vertex is

$$v = \frac{4136912499509}{2748779069440} + i\frac{92225107213}{549755813888}.$$

Control	q_P	One-step rate	Exact power	C_0	C_2
Normal diag$(1, 2)$	0.333333	17	17	18	18
$I_{32} + \frac{1}{5}N_{32}$	0.216478	12	12	13	13
Shifted sharp $I_2 + J$	1.082392	—	2	—	—
Coupled family, $n = 32$	0.532135	113	11	31	31

Table 1: Sufficient relative residual budgets for $\tau = 10^{-8}$. Here $C_0 = 1 + \sqrt{2}$ is implemented with a rational upper bound, and the $C_2 = 2$ column uses the unconditional numerical-radius power inequality for Richardson powers. The one-step rate uses $(\|E\|_1\|E\|_\infty)^{m/2}$. The exact-power column uses $\|E^m\|_1\|E^m\|_\infty \le \tau^2$. Both norm certificates have constant one. Decimals are descriptive; stopping decisions use exact rational inequalities. Dashes mean that the corresponding geometric or norm-rate certificate does not certify contraction. Exact powers certify the shifted sharp control in two steps. The table compares sufficient certificate budgets.

The other is its complex conjugate. Using $C_{\mathrm{up}} = 663613970941/274877906944 \ge 1 + \sqrt{2}$, the exact test (6.2) gives 31 steps. The same test with $C_{\mathrm{up}} = 2$ from the unconditional numerical-radius power inequality also gives 31 steps. Both 30-step tests fail. A separate rational recomputation of all supporting row bounds and feasible line intersections reproduced (7.2). The reduction relative to the single-step 113-iteration rate budget comes from enclosure geometry. Direct powers provide a stronger competing certificate, as quantified below.

Stronger exact-power certificate. For this same $n = 32$ input, exact rational powers of $E = I - A$ give

$$\|E^{11}\|_1\|E^{11}\|_\infty = \frac{505768790783}{24414062500000000000000000000} < 10^{-16}. \tag{7.3}$$

Therefore 11 iterations suffice for the relative target 10^{-8} under this standard norm-product certificate. At $m = 10$, the corresponding product is

$$\frac{217552976919}{24414062500000000000000000000} > 10^{-16}.$$

An exact scan of powers $1, \dots, 11$ first passes at 11. These statements were checked by a separately implemented dense rational matrix multiplication. The package reference baseline also uses generic exact matrix multiplication. Failure of the 10-step upper-bound test does not prove that an actual residual cannot reach the target earlier. The 11-step power certificate beats both numerical-range budgets of 31. This fixture demonstrates recomputability and the conservatism of the single-step baseline, while direct powers supply the strongest displayed certificate.

The controls in Table 1 test distinct properties: degenerate geometry, a cheap dominating bound, sharpness with unsuccessful geometric contraction certification, and a weak one-step baseline exposed by exact powers. Where numerical-range budgets are available, the strongest displayed norm certificate matches or improves them. The cost study below measures construction, checking and direct powers on specified analytic fixtures and discretization models. Validated spectral-norm methods and block-power reuse provide additional comparators for a broader performance study. Certificate selection depends on both bound quality and construction cost.

7.4 A scalar enclosure and mesh error study

To exercise the general-polynomial mesh certificate, rather than the Richardson vertex shortcut, consider

$$A = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad A^2 = I.$$

For an integer normal (a, b), the support Hermitian matrix has eigenvalues $\pm\frac{1}{2}\sqrt{5a^2 + b^2}$. Thus its numerical range is the ellipse $x^2/(5/4)+y^2/(1/4) \le 1$. The polynomials $p_1(z) = z^2$ and $p_2(z) = 1-z^2$ both have $M_p = \max_{W(A)} |p| = 5/4$, whereas $\|p_1(A)\| = 1$ and $\|p_2(A)\| = 0$. For the second scalar maximum, write the ellipse boundary as $z = (\sqrt{5}\cos\theta + i\sin\theta)/2$. Then

$$|1 - z^2|^2 = 25/16 - (5/2)t + t^2, \qquad t = \cos^2\theta \in [0, 1].$$

The convex quadratic has maximum $25/16$ at $t = 0$. The zero matrix polynomial is a deliberate negative control: a scalar supremum certificate need not detect annihilation at the matrix.

The study compares the shipped Gershgorin support construction with an experimental analytic support producer that rounds the displayed support outward. This experimental producer exercises the support-accuracy interface; the shipped backend uses Gershgorin bounds. We cross resolutions $k = 1, 2, 4, 8$, square-root precisions $b = 8, 24, 40$, and $s = 4, 16, 64$ segments per edge, for both producers and polynomials: 144 cases in total. If $\bar{h}_q$ is the unnormalized produced support, set

$$R_b = \mathrm{sqrt_upper}(5/4, b), \qquad \eta = \max_q \frac{\bar{h}_q - \max\{0, \mathrm{sqrt_upper}((5a^2 + b_q^2)/4, b) - 2^{-b}\}}{k},$$

where $q = (a, b_q)$ distinguishes a normal coordinate from the bit precision. All normals have length at least k, and their angular gap satisfies $\Delta \le \arctan(1/k) \le 1/k$. Consequently the following rational relaxations of Proposition 3.2 suffice:

$$\rho = \frac{R_b + \eta}{1 - 1/(8k^2)}, \qquad d_P = \eta + \frac{R_b + \rho}{2k}, \qquad D_p = 2\rho.$$

They use $\cos x \ge 1 - x^2/2$ and $\sin x \le x$, avoiding unchecked trigonometric evaluations. The mesh radius δ is an outward upper bound for the longest edge divided by $2s$. Exact rational assertions check every vertex radius and all 144 inequalities

$$0 \le U_p - 5/4 \le 2\rho(d_P + \delta) + 2^{-b}.$$

The CSV preserves the exact rationals for each term; decimal columns and Table 2 are descriptive only.

For analytic supports, $\eta \le 2^{-b}/k$; finer directions and meshes reduce the observed excess. The Gershgorin polygon contains $\pm 3/2$ for every displayed resolution: its support is $|a| + \frac{1}{2}\sqrt{a^2 + b_q^2}$, which dominates the support of those points. Hence its z^2 supremum is at least $9/4$, leaving an irreducible excess of at least one over the true $5/4$, even with arbitrarily many directions. Mesh refinement alone cannot correct support slack. The producer-based experiment uses shared geometry helpers and checks the finite error accounting. The accompanying independent arithmetic oracle recomputes this study's exact fields using separate root, clipping, and polynomial-evaluation implementations.

Support producer	k	U_{z^2}	U_{1-z^2}	Excess budget
Analytic	1	1.290326	1.278939	3.090163
Analytic	2	1.271082	1.268033	1.329167
Analytic	4	1.261034	1.260257	0.642659
Analytic	8	1.255562	1.255367	0.318785
Gershgorin	1	2.361888	1.426620	7.223737
Gershgorin	2	2.321651	1.343508	3.939560
Gershgorin	4	2.307298	1.313327	2.710893
Gershgorin	8	2.302639	1.304187	2.144144

Table 2: Scalar mesh bounds at 40 root bits and 64 segments per edge. Both true scalar suprema are 5/4. The last column bounds $U_p - M_p$, with the same conservative budget for both polynomials. These are scalar supremum bounds before applying a matrix theorem constant.

7.5 Measured certificate costs

We predefine ten fixtures: the five comparison controls, the $n = 2, t = 99/100$ nilpotent counterexample, the coupled family at $n = 8$, and three centered-difference matrices for $-\nu u'' + vu' + ru$ on $(0, 1)$ with homogeneous Dirichlet boundaries. The latter use $\nu = 1/100$, $v = 1/10$, $r = 1$, $h = 1/(n + 1)$, $n = 8, 16, 32$, diagonal $d = 1 + 2\nu/h^2$, lower diagonal $-\nu/h^2 - v/(2h)$, and upper diagonal $-\nu/h^2 + v/(2h)$. We set $\alpha = 1/d$. These specified mathematical models supply reproducible discretization fixtures for the certificate-cost comparison.

Every request uses eight directions, 40 root bits, relative tolerance 10^{-8}, and the unconditional Richardson constant two. Three sequential observations time each complete producer call, checker call, and dense exact-power search, after one untimed 1×1 producer warmup. Construction includes the one-step baseline; checking includes full regeneration and JSON evidence comparison. The direct-power search checks every smaller positive power up to the first certificate or the fixed cap 64. A separate norm-only call normalizes the request, constructs $I - \alpha A$, computes its one-step norm product and searches its budget, using shared arithmetic helpers while skipping all support geometry. Serialization time is excluded from construction; compact UTF-8 JSON sizes measure evidence size, not peak memory. The recorded environment is Python 3.11.5 on Windows; `results/certificate-cost-study.json` stores its full platform, all observations and source hash. Timings are environment-specific medians of certificate construction, checking, and reference calls.

For the coupled $n = 32$ fixture, the exact-power certificate is both stronger and cheaper to construct than the numerical-range certificate in these observations. All three discretization cases are already certified by the cheaper one-step norm route with smaller budgets: 39 versus 41, 116 versus 121, and 411 versus 427, respectively. The norm-only timing separates that route's work from unnecessary support construction. At sizes 16 and 32 the direct-power search is inconclusive at its cap; this does not make numerical-range geometry necessary for certification. Longer or block-power searches could change both cost and bound quality. The successful norm-based comparators account for the displayed ordering of certificate budgets and construction costs. The numerical-range certificate evidence ranges from 1647 to 15263 bytes on these fixtures. The measured quantities are certificate timings and evidence sizes; solver runtime, residual histories, peak memory, and validated spectral-norm comparisons require a broader study.

Fixture	NR	One	Power	Build	Check	Norm	Power
Normal	18	17	17	1.05	1.07	0.08	0.45
Nilpotent $n = 32, t = 1/5$	13	12	12	85.78	85.90	7.14	69.58
Coupled $n = 32$	31	113	11	84.36	85.10	7.16	72.92
Shifted sharp	—	—	2	1.17	1.16	0.05	0.08
Slow scalar	1902	1833	cap	3.98	3.96	1.47	0.86
Nilpotent $n = 2, t = 99/100$	31	1833	2	2.75	2.75	1.49	0.08
Coupled $n = 8$	31	113	11	6.20	6.24	0.54	4.12
Discretization $n = 8$	41	39	29	6.28	6.29	0.53	12.73
Discretization $n = 16$	121	116	cap	22.50	22.73	1.96	133.98
Discretization $n = 32$	427	411	cap	88.32	90.38	7.75	738.15

Table 3: Three-observation median certificate timings. NR and One denote numerical-range and one-step iteration budgets; the first Power column is an iteration budget. Build, Check, Norm and the last Power column are median milliseconds. "cap" is an unsuccessful 64-power run and must not be ranked as a successful competing certificate. A dash indicates absence of a geometric contraction certificate.

8 Limitations, reproducibility, and research use

The results certify polynomial operators for the stated exact inputs. Physical model adequacy and finite-precision solver behavior require their corresponding model and roundoff analyses. Certificate selection depends on bound strength and construction cost. On the featured fixture, the standard exact-power certificate is stronger than either numerical-range mode. Coefficient growth can make the derivative estimate large. A basis other than shifted monomials may reduce this effect, but requires a corresponding proved derivative bound.

The exact rational scalar prototype is restricted. Sparse validated eigenvalue methods, scalable half-plane intersection, general matrix-coefficient sampling, adaptive meshes, and a verified floating-point solver interface are further engineering tasks. The enclosure excess estimate requires validated support accuracy if it is used as a quantitative convergence bound. Without that evidence, the main certificate can remain valid but conservative.

The analytic controls are completely specified in Section 7; they can be independently reconstructed without repository code or a random seed. The accompanying `nrcert` version 0.1.0 source bundle includes the exact requests, generated certificates, a 30-row comparison CSV, and command logs. A wheel installed into a separate Python 3.11.5 environment passed all 44 tests, including evidence mutation and recomputation tests, and all six command-line examples. The commands are reproduced by `reproduce/verify_package.py` in the complete source distribution, which also contains the test suite; the environment and exit statuses are recorded in `audit/installed-package-verification.json`. Producer and checker share an implementation. The checker recomputes the complete normalized request and compares regenerated evidence, providing reproducibility and tamper detection relative to that implementation. Separate arithmetic validation is supplied by the independent ellipse oracle for its specified finite study. Together, the exact fixtures, regeneration checks, and recorded comparisons support the prototype's certificate specification and error accounting. Operational limits for this version are dimension 64, degree 128,

direction resolution 8 (64 normals), 128 mesh segments per edge, iteration search cap 100000, and square-root precision 8–256 bits. The reference power routine permits at most 128 powers; this study uses 64. Raw rational strings and derived factors have 8192-character limits, with 8192-bit raw and 12000-bit derived numerator/denominator budgets. Compact decimal exponents are checked before conversion, including supported Unicode decimal digits. Matrix dimension, degree, mesh size and iteration search are capped; exact arithmetic can still be expensive within those caps. Factored witnesses avoid output expansion; power-comparison costs still depend on exponents and rational operand sizes. Malformed or over-budget input raises the documented error. The interface supports research computation under these resource limits. Exploratory floating-point norms are recorded as diagnostics.

Generative AI disclosure

OpenAI Codex agents contributed substantially to source discovery, literature synthesis, and mathematical exposition.

References

[1] Michel Crouzeix and César Palencia. The numerical range is a $(1+\sqrt{2})$-spectral set. *SIAM Journal on Matrix Analysis and Applications*, 38(2):649–655, 2017. doi: 10.1137/17M1116672. URL https://epubs.siam.org/doi/10.1137/17M1116672.

[2] Bernard Delyon and François Delyon. Generalization of von Neumann's spectral sets and integral representation of operators. *Bulletin de la Société Mathématique de France*, 127(1):25–41, 1999. doi: 10.24033/bsmf.2340. URL https://smf.emath.fr/node/42955.

[3] Catalin Badea, Michel Crouzeix, and Bernard Delyon. Convex domains and K-spectral sets. *Mathematische Zeitschrift*, 252:345–365, 2006. doi: 10.1007/s00209-005-0857-y. URL https://link.springer.com/article/10.1007/s00209-005-0857-y.

[4] Michel Crouzeix. Numerical range and functional calculus in Hilbert space. *Journal of Functional Analysis*, 244(2):668–690, 2007. doi: 10.1016/j.jfa.2006.10.013. URL https://www.sciencedirect.com/science/article/pii/S002212360600440X.

[5] Charles R. Johnson. Numerical determination of the field of values of a general complex matrix. *SIAM Journal on Numerical Analysis*, 15(3):595–602, 1978. doi: 10.1137/0715039. URL https://epubs.siam.org/doi/10.1137/0715039.

[6] Sébastien Loisel and Peter Maxwell. Path-following method to determine the field of values of a matrix with high accuracy. *SIAM Journal on Matrix Analysis and Applications*, 39(4):1726–1749, 2018. doi: 10.1137/17M1148608. URL https://epubs.siam.org/doi/10.1137/17M1148608.

[7] L. N. Trefethen. Field of values and numerical abscissa. Chebfun example, November 2010, 2010. URL https://www.chebfun.org/examples/linalg/FieldOfValues.html. Accessed 7 October 2026.

[8] Bernhard Beckermann and Lothar Reichel. Error estimates and evaluation of matrix functions via the Faber transform. *SIAM Journal on Numerical Analysis*, 47(5):3849–3883, 2009. doi: 10.1137/080741744. URL https://epubs.siam.org/doi/10.1137/080741744.

[9] Michel Crouzeix and Anne Greenbaum. Spectral sets: Numerical range and beyond. *SIAM Journal on Matrix Analysis and Applications*, 40(3):1087–1101, 2019. doi: 10.1137/18M1198417. URL https://epubs.siam.org/doi/10.1137/18M1198417.

[10] Youcef Saad and Martin H. Schultz. GMRES: A generalized minimal residual algorithm for solving nonsymmetric linear systems. *SIAM Journal on Scientific and Statistical Computing*, 7(3):856–869, 1986. doi: 10.1137/0907058. URL https://doi.org/10.1137/0907058.

[11] Michael Eiermann and Oliver G. Ernst. Geometric aspects of the theory of Krylov subspace methods. *Acta Numerica*, 10:251–312, 2001. doi: 10.1017/S0962492901000046. URL https://www.cambridge.org/core/journals/acta-numerica/article/abs/geometric-aspects-of-the-theory-of-krylov-subspace-methods/5D5142ABA62629A62A8272F9F59B6F26.

[12] Bernhard Beckermann. Image numérique, GMRES et polynômes de Faber. *Comptes Rendus Mathematique*, 340(11):855–860, 2005. doi: 10.1016/j.crma.2005.04.027. URL https://www.sciencedirect.com/science/article/pii/S1631073X05001755.

[13] Anne Greenbaum, Vlastimil Pták, and Zdeněk Strakoš. Any nonincreasing convergence curve is possible for GMRES. *SIAM Journal on Matrix Analysis and Applications*, 17(3):465–469, 1996. doi: 10.1137/S0895479894275030. URL https://epubs.siam.org/doi/10.1137/S0895479894275030.

[14] Emiel Lorist and Felix L. Schwenninger. A solution to Crouzeix's conjecture. arXiv preprint, 2026. URL https://arxiv.org/abs/2608.03841v2. Version 2, 17 August 2026.

[15] Michel Crouzeix and Anne Greenbaum. Update to: Spectral sets: Numerical range and beyond. arXiv preprint, 2026. URL https://arxiv.org/abs/2609.22460v2. Version 2, 23 September 2026.

[16] Michel Crouzeix and Anne Greenbaum. A new proof that the numerical range is a complete 2-spectral set for weighted shift matrices. arXiv preprint, 2025. URL https://arxiv.org/abs/2508.12768v1. Version 1, 18 August 2025.

[17] Per Åhag, Rafał Czyż, and Jani Virtanen. The complete Crouzeix conjecture in dimension three and the Clouâtre–Ostermann–Ransford conjecture. arXiv preprint, 2026. URL https://arxiv.org/abs/2608.27346v4. Version 4, 21 September 2026.

[18] OpenAI. A direct proof of the complete Crouzeix inequality. OpenAI Math Release preprint OAI:A-direct-proof-of-the-complete-Crouzeix-inequality-September-26-2026, 2026. URL https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-direct-proof-of-the-complete-Crouzeix-inequality-September-26-2026/paper.pdf. Manuscript dated 26 September 2026; audited repository snapshot adc7f1241b42e322a6451854ab7e4b4c146bf78a, accessed 7 October 2026.

[19] Nicholas J. Higham. *Functions of Matrices: Theory and Computation*. Society for Industrial and Applied Mathematics, Philadelphia, 2008. doi: 10.1137/1.9780898717778. URL https://epubs.siam.org/doi/10.1137/1.9780898717778.

[20] Marlis Hochbruck and Alexander Ostermann. Exponential integrators. *Acta Numerica*, 19:209–286, 2010. doi: 10.1017/S0962492910000048. URL https://doi.org/10.1017/S0962492910000048.

[21] Awad H. Al-Mohy and Nicholas J. Higham. Computing the action of the matrix exponential, with an application to exponential integrators. *SIAM Journal on Scientific Computing*, 33(2):488–511, 2011. doi: 10.1137/100788860. URL https://epubs.siam.org/doi/10.1137/100788860.

[22] OpenAI. Mathematical manuscripts and supporting proof artifacts: repository README. GitHub repository, 2026. URL https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/README.md. Snapshot adc7f1241b42e322a6451854ab7e4b4c146bf78a; accessed 7 October 2026.

[23] OpenAI. The complete Crouzeix conjecture: Lean formalization scope, family 325. OpenAI Math Release, Lean scope document, 2026. URL https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/lean/docs/325.md. Scope statement only; not an independent compilation report; accessed 7 October 2026.

[24] Siegfried M. Rump. Verification methods: Rigorous results using floating-point arithmetic. *Acta Numerica*, 19:287–449, 2010. doi: 10.1017/S096249291000005X. URL https://doi.org/10.1017/S096249291000005X.

[25] Siegfried M. Rump. Verification of positive definiteness. *BIT Numerical Mathematics*, 46(2):433–452, 2006. doi: 10.1007/s10543-006-0056-1. URL https://link.springer.com/article/10.1007/s10543-006-0056-1.

[26] Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. Society for Industrial and Applied Mathematics, Philadelphia, 2 edition, 2002. doi: 10.1137/1.9780898718027. URL https://epubs.siam.org/doi/book/10.1137/1.9780898718027.

[27] Eitan Tadmor. On the stability of Runge–Kutta methods for arbitrarily large systems of ODEs. *Communications on Pure and Applied Mathematics*, 78(4):821–855, 2025. doi: 10.1002/cpa.22238. URL https://doi.org/10.1002/cpa.22238.

[28] A. S. Lewis and M. L. Overton. Partial smoothness of the numerical radius at matrices whose fields of values are disks. *SIAM Journal on Matrix Analysis and Applications*, 41(3):1004–1032, 2020. doi: 10.1137/18M1236289. URL https://epubs.siam.org/doi/10.1137/18M1236289.

[29] Valeria Simoncini and Daniel B. Szyld. Theory of inexact Krylov subspace methods and applications to scientific computing. *SIAM Journal on Scientific Computing*, 25(2):454–477, 2003. doi: 10.1137/S1064827502406415. URL https://epubs.siam.org/doi/10.1137/S1064827502406415.

[30] Nicholas J. Higham and Françoise Tisseur. A block algorithm for matrix 1-norm estimation, with an application to 1-norm pseudospectra. *SIAM Journal on Matrix Analysis and Applications*, 21(4):1185–1201, 2000. doi: 10.1137/S0895479899356080. URL https://epubs.siam.org/doi/10.1137/S0895479899356080.